

XAI-relevant EWS design guidelines: evidence catalogue

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50 of the 128 EWS design guidelines affect how a flood prediction is explained: 31 from the triage in the Research Approach and 19 added from Minindu's manual review of the guideline catalogue. Each one is grouped by how much published evidence links it to understanding explanations, forecasts or warnings.

Evidence at a glance

EVIDENCE	FROM TRIAGE	NEW	TOTAL
Direct	26	12	38
Indirect	4	5	9
None found	1	2	3

Guidelines in each group

Direct	G4, G11, G14, G16, G18, G19, G21, G22, G25, G26, G41, G47, G48, G50, G54, G57, G58, G59, G67, G73, G76, G77, G79, G80, G81, G88, G89, G107, G111, G114, G117, G118, G119, G120, G121, G124, G125, G128
Indirect	G15, G17, G23, G24, G27, G38, G49, G84, G109
None found	G20, G42, G65

How to read this document

- Direct** A source shows the property changes how people understand an AI explanation, a probability or forecast, or a warning.
- Indirect** A source shows it affects comprehension in general text or interfaces, or in people with cognitive impairments. Applying it to explanations is our inference.
- None found** Marked as possibly relevant, but no source links it to understanding. Kept as our inference.
- B · XAI** changes what is explained or how it is understood. **C · Both** a UI guideline that also shapes the explanation.
- Research Approach v2** from the original triage. **New, manual marking** added from the colour-coded catalogue. A red note shows where the catalogue marking differs.
- Where the catalogue marking and the Research Approach v2 triage gave different classes (G11, G41, G49, G67, G84, G107, G117, G121, G124), the v2 class is kept.
- Each source says how it was checked: **full text** or **abstract**. Nothing rests on a search snippet. "Counter" marks evidence against the guideline.

Limits

No source tests these guidelines on AI explanations for people with dementia, MCI, autism or ADHD, so every link to that population is our inference. None of the 49 catalogue sources discusses AI explanations. Publisher sites were blocked during the search, so most sources were checked at abstract level.

Why standard XAI output needs this adaptation at all

- Miller (2019), Artificial Intelligence 267. It is "not reasonable to expect a lay-user to be able to interpret a causal chain"; explanations are social and fitted to what the listener knows. **full text**
- Liao, Gruen & S. Miller (2020), CHI. Lay users ask for explanations; practitioners found LIME feature-importance lists "unsatisfying". **full text**
- Schuff et al. (2022), FAccT. Lay users "often mis-interpret" feature-attribution explanations, including Shapley-based ones. **full text**
- Tielman et al. (2024), Technology in Society 79. "Adaptive XAI is the key to inclusive XAI"; it calls for personalisation, "more than just simplicity in text". **full text**

Direct 38 guidelines

A source shows this property changes how people understand an AI explanation, a probability or forecast, or a warning message.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G14	Use clear words B · XAI New, manual marking	Raw XAI output uses feature names (<code>soil_moisture_72h</code> , "probability"). Each reason must become everyday words ("the ground is already wet").	A lookup table in the data model maps each model input to plain words in Sinhala, Tamil and English. The "Why?" screen reads only from it.	Bullock et al. (2019), Public Underst. Sci. Jargon "impairs people's ability to process scientific information" (N=650). abstract Walpole et al. (2025), Risk Analysis. "Evacuation warning" was poorly understood in emergency alerts; omit jargon where possible. abstract Zytek et al. (2022). "ML models are only as interpretable as their features"; features must be readable by users. full text Gigerenzer et al. (2005), Risk Analysis. "30% chance of rain" is read in contradictory ways unless the reference class is stated. abstract
G16	Avoid double negatives or nested clauses B · XAI New, manual marking	Uncertainty wording invites negatives ("it is not unlikely"). One clause per sentence; no negated likelihood words.	A banned-phrase list in the template rules; Layer 4 AI wording that contains one is rejected.	Teigen & Brun (1995), Acta Psychologica. "Unlikely" and "doubtful" "function more like negations". abstract Teigen & Brun (1999), OBHDP. Phrases of equal probability lead to different decisions depending on direction. abstract Budescu et al. (2009), Psychological Science. Readers' numeric readings of IPCC probability words "deviated significantly" from the guidelines. abstract Xiang et al. (2020): negative sentences cost more processing. Nasiri et al. (2022): mild Alzheimer's impairs relative-clause sentences. abstract "Not unlikely"-style wording itself is untested; that step is our inference.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G18	Keep text succinct B · XAI New, manual marking	Each explanation step is one sentence of 15 words or fewer; the five invariants fit in three sentences.	A word budget per text slot in each screen spec, so the forecast card never scrolls.	mixed evidence Narayanan et al. (2018), preprint. Longer, more complex ML explanations raised response time and lowered satisfaction; accuracy stayed stable. summary van Boxtel et al. (2021). Comprehension of long sentences declines with ageing and more with Alzheimer's. abstract Counter: Carlson et al. (2024). 360-character alerts did not beat 90-character ones on compliance; shortness must not drop key content. abstract
G19	Use clear formatting and punctuation B · XAI New, manual marking	No all-caps or "!!!" in explanations (v2 says "Danger! Flood coming!"). Numbers written one consistent way ("3 in 10", "30%").	Style rules for the forecast card and the likelihood component.	mixed evidence Waugh et al. (2025), Weather, Climate & Society. Emphasising key text in a warning (bold or caps) aids attention and memory. Supports selective emphasis, not all-caps blocks. abstract Klinke et al. (2024). All-uppercase text slows reading and hurts readability of long headlines. abstract Counter: Arditi & Cho (2007), Vision Research. Uppercase was as legible or more at small sizes. abstract
G21	Provide summary of long documents Need: Get the gist first B · XAI Not marked in Excel Research Approach v2	Open with a one-sentence bottom line: what, where, when, what to do	The forecast card and the push notification both start with the same bottom-line sentence; reasons and evidence sit behind "Why?" and "How sure?"	Schwartz, Woloshin & Welch (2009), Ann Intern Med. Replacing a long summary with a one-table facts box raised correct understanding of benefit from 9% to 72% (two RCTs). abstract Peixoto et al. (2025), preprint. Users said simplified LIME/SHAP explanations and an overview gave better quick understanding (n=3). full text Nathe & Krakow (2019): patient understanding fell as page count rose. abstract No study compares bottom line first vs later.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G22	Separate each instruction B · XAI New, manual marking	The action part of the explanation ("get your go-bag ready") is a numbered list, one step per line.	Action list with a checkbox per step under the forecast card, fed by the CAP instruction field.	mixed evidence Desaulniers (1987), <i>HFS Proc.</i> Outline-format warnings were read and followed more than paragraph warnings. abstract Morrow et al. (1998), <i>Exp. Aging Res.</i> List instructions were understood and recalled better and reduced age and working-memory differences. abstract Counter: Frantz (1993), <i>J. Safety Research</i>. Prose vs numbered list did not change label effectiveness. abstract
G25	Explain implied content Need: Understand what the prediction means B · XAI Research Approach v2	Explain what a risk category means in concrete consequences, not the label alone	Each app state (Safe, Forecast, Warning, Danger) stores a consequence line shown under its badge, e.g. "Water may come into houses near the river"	mixed evidence Weyrich et al. (2018), <i>Weather, Climate & Society</i>. Impact information plus recommendations improved understanding of the warning and intended response. abstract Potter et al. (2018), <i>IJDRR</i>. Impact-based warnings increased understanding of potential impacts (n=1,364). abstract Counter: Potter et al. found understanding did not raise action; Choi et al. (2025): impact-based heat warnings did not change intentions. abstract
G26	Alternatives for numerical concepts Need: Understand likelihood without numeracy B · XAI Research Approach v2	Word + picture (e.g. 3 of 10 houses), exact number on request	One likelihood component renders the profile's form: word card, icon array of 10 houses, "30%" on tap; reused on the card, the notification and the "How sure?" screen	mixed evidence Galesic et al. (2009), <i>Health Psychology</i>. Icon arrays "increased accuracy of both low- and high-numeracy people", including adults aged 62–77. abstract Gigerenzer & Hoffrage (1995), <i>Psychol Rev</i>: frequency formats improved reasoning. McDowell & Jacobs (2017), <i>meta-analysis</i>: natural frequencies and visual aids help. abstract Counter: Galesic et al.: risks shown as icon arrays seemed "less serious"; matters for warnings. abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G47	Don't rely on user calculation Need: Not work things out from numbers B · XAI Not marked in Excel Research Approach v2	"In 2 days" becomes "Thursday morning"; the system compares forecasts for the user	The app converts lead time to day and part of day, and computes the change from the previous forecast; the user never subtracts dates or compares percentages	Peters et al. (2007), Med Care Res Rev. Formats that "ease the cognitive burden" gave higher comprehension, most for low-numeracy users. abstract Liu et al. (2019), Aging Med: calculation and orientation to time are among the most impaired items in MCI and dementia (n=4,487). Boyle et al. (2013): cognitive decline predicts lower numeracy. abstract
G50	State results of actions Need: Know what it means for their actions B · XAI Research Approach v2	State what is likely to happen if they stay vs leave	Action card under the forecast gives one line for staying and one for leaving, linked to the CAP instruction field	Sutton, Fischer & Wood (2021), Weather, Climate & Society. Protective action guidance increased understanding and the ability to decide what to do. abstract Fischer et al. (2023). Guidance in text and graphic increased understanding and deciding across hazards. abstract These test what to do, not an explicit stay-vs-leave comparison.
G59	Info and Relationships (SC 1.3.1) B · XAI New, manual marking	The explanation's structure (bottom line, reason, how sure) is exposed to screen readers. The icon array needs a text equivalent.	accessibilityRole="header" on each step; icon array labelled "3 of 10 houses filled".	Sharif et al. (2021), ASSETS. Screen-reader users were 61% less accurate and took 211% longer with online charts; they asked for text and table versions. abstract Zong et al. (2022), Comput. Graph. Forum. Structured, described charts helped blind readers understand data. abstract Evidence is for visual impairment, not cognitive impairment. WCAG 1.3.1 text not re-checked.
G73	Persistent visual context Need: Know where and when it applies B · XAI Not marked in Excel Research Approach v2	Always show the prediction's area and time window	A fixed header on every forecast and explanation screen: area name, time window and the "Synthetic data" label	Klockow-McClain et al. (2020), Ann Am Assoc Geogr. Being inside or outside the warning area and distance changed risk estimates (N=5,564). abstract Yoon et al. (2024), JMIR: location was the only alert content linked to people leaving the area. Peters-Founshtein et al. (2023), preprint: impaired spatio-temporal orientation in MCI. abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G89	Warning before any change Need: Not be surprised by a revised forecast B · XAI Not marked in Excel Research Approach v2	Announce and explain an upgrade or downgrade before the new value	When a new forecast changes the level, a "The forecast has changed" banner and notification appear before the new value, with a one-line reason	mixed evidence Burgeno & Joslyn (2020) . Inconsistency between successive forecasts reduced trust. Losee & Joslyn (2018) : inconsistency lowered trust but increased protective action. abstract Counter: Burgeno et al. (2023) : inconsistency increased trust with probabilistic forecasts. Baseman et al. (2013) : each extra alert per week cut recall of content by 41%. abstract No study tests announcing a change before the new value.
G118	Multi-modal alert (audio + visual + haptic) B · XAI New, manual marking	A forecast change or upgrade is announced through more than one channel. Overlaps G128.	Forecast-change notification uses text, a voice line and a short vibration.	Stephens et al. (2013) , Hum. Commun. Res. Three redundant messages, one through a synchronous channel, gave the fastest grasp of urgency (N=1,318). abstract Tannenbaum-Baruchi et al. (2026) , Disability & Health J. Sound + vibration + light app alerts; authors call for multimodal systems. Maguire et al. (2025) , IJDRS : warnings in "multiple formats and channels". abstract Both disability studies are with Deaf and hard-of-hearing adults; none with cognitive impairment.
G119	CAP message structure Need: Get likelihood, severity, urgency in a standard form B · XAI Research Approach v2	CAP's certainty field (Observed / Likely / Possible / Unlikely) gives a standard word for likelihood	The forecast data model carries CAP fields (certainty, severity, urgency, instruction); the certainty word drives the likelihood word on every screen	mixed evidence Taylor et al. (2023) , Risk Analysis . Likelihood words in warnings changed judgments; "unlikely" lowered perceived risk and action even for severe impacts. abstract Budescu et al. (2009) , Psychological Science . Readings of probability words deviated from official meanings; pair words with numbers. abstract Counter to bare CAP words: supports adding the picture or number. OASIS CAP 1.2 text not re-checked.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G120	Universal hazard symbols B · XAI New, manual marking	The hazard invariant always carries the standard flood pictogram, so the hazard is clear without reading.	One fixed flood symbol on every forecast, explanation and notification screen.	Shinar et al. (2013), <i>Transp. Res. F</i>. Adding text to a symbol improves comprehension and speed, especially for unfamiliar signs. abstract Shorr et al. (2009), <i>HFES</i>. Older adults understood ANSI warning symbols less accurately and more slowly. abstract Boedeker et al. (2023), <i>J Gerontol B</i>. Symbols with text cut errors for people with Alzheimer's dementia. abstract
G4	One step per screen C · Both Not marked in Excel Research Approach v2	Stage the explanation: bottom line, reason, evidence, uncertainty	"Why?" opens a stepper: bottom line, reason, evidence, how sure, one screen each with Back and Next	mixed evidence Mindlin et al. (2024), <i>preprint</i>. Lay users who explored the explanation one question at a time understood the model significantly better than with a static all-at-once report (n=107); recommends layering "while not overwhelming the users with all information at once". full text Bahel, Sriram & Conati (2024). Explanations split over pages opened one at a time raised engagement, understanding and learning. full text Counter: Mindlin et al. found no difference in self-rated understanding or confidence.
G11	Most important stands out C · Both Research Approach v2	Lead with conclusion and action; the "why" is secondary	Bottom line and action button fill the top of the forecast card; "Why?" is a smaller secondary button below	Peters et al. (2007), <i>Med Care Res Rev</i>. "Less is more": formats that highlight the meaning of important information gave higher comprehension, most for low-numeracy readers. abstract Bahel et al. (2024). Showing the explanation more prominently than the conclusion confused users; making the conclusion dominant fixed it (pilot). full text Mathur et al. (2026), <i>CHI</i>. Older adults used AI explanations to calibrate urgency and decide on action. full text

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G41	Avoid too much content C · Both Research Approach v2	Progressive disclosure; detail level is a profile attribute	The profile's "detail on first view" setting decides how many explanation parts render; the rest open on tap	Peters et al. (2007). Less information, focused on meaning, improved comprehension. abstract Lage et al. (2019), preprint. Explanation complexity reduced human interpretability; new concepts cost more than repetition. abstract Waterschoot et al. (2025), UMAP. More detailed explanations did not increase understanding (RCT, n=271). Monnet et al. (2024): low cognitive load is a key need in dementia decision tools. abstract
G48	Provide human help C · Both Not marked in Excel Research Approach v2	"Ask someone to explain" as an explanation option	An "Ask someone to explain" button on explanation screens calls or messages the saved helper or the emergency hotline	mixed evidence Álvarez-Aguado et al. (2026), J Intellect Disabil Res. Adults with intellectual disability "rejected AI as a substitute for human support"; AI should complement their support network. abstract Wilson et al. (2026), JARID. People with intellectual disability valued Easy Read health information "and assistance in appraising it". abstract Counter: Buell et al. (2019), JARID. Human mediation alone did not significantly improve understanding (RCT, n=60); simplified text + mediation did best post hoc. abstract
G54	Provide reminders C · Both Not marked in Excel Research Approach v2	Reminders re-explain the current prediction and flag changes	Scheduled local notifications restate the current forecast before its time window and say if it has changed	mixed evidence Herdener et al. (2018), Human Factors. Earlier hurricane forecasts anchored judgments, and cognitive load impaired adjusting to new information; supports flagging changes. abstract Hackett et al. (2025): smartphone prompts raised task completion in MCI and mild dementia (n=10). abstract Mixed: Ghani et al. (2026), JMIR. A reminder app for MCI gave only modest quality-of-life gains (RCT, n=1,078). abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G57	Support simplification C · Both Not marked in Excel Research Approach v2	Simple mode = minimal explanation that still keeps the invariants	Simple mode shows one card with the five invariants only; history, map and reasons are hidden	mixed evidence Lage et al. (2019). Simpler explanations were more interpretable. abstract Peixoto et al. (2025), preprint. Simplified LIME/SHAP explanations were more comprehensible for non-visual users (n=3, preliminary). full text Counter: Xuan et al. (2023), IJHCS. Highly comprehensible explanations were most often misread, with overconfident inferences (n=200). Simple mode must keep the key facts. abstract
G58	Personalised, persistent interface C · Both Not marked in Excel Research Approach v2	Explanation profile persists and stays editable	The profile is stored on the device, loaded at start-up and edited in Settings under "How we explain"	Bahel, Sriram & Conati (2024). Explanations personalised to user traits significantly increased engagement, understanding and learning. full text Tielman et al. (2024). "Adaptive XAI is the key to inclusive XAI" (position paper). full text No source tests whether the profile should persist or be editable.
G67	Reading level C · Both Research Approach v2	Explanation meets a target reading level; fuller version as supplement	Explanation templates are written to the target reading level; a "More detail" link opens the fuller version	Walpole et al. (2025), Risk Analysis. "Evacuation warning" was poorly understood in emergency alerts across experience levels. abstract Mikhail et al. (2015), RCT. Grade-5 pamphlets were understood better than grade-10 (57.9% vs 48.3%), whatever the reader's literacy. abstract
G76	Realistic, non-childish imagery C · Both New, manual marking	The landmark comparison uses a real photo of the local bridge marker, not a cartoon.	Per-gauge landmark photos stored with the local reference levels (already a capability in Task 5).	Wilhelmi et al. (2024), IJDRR. Realistic images of flooded buildings helped residents understand storm-surge forecasts. abstract Zannino et al. (2007), (2010). Photos beat line drawings for naming in Alzheimer's. abstract The "non-childish" part is untested.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G77	Avoid metaphorical icons C · Both New, manual marking	"How sure?" uses no dice or gauge icon; reasons use literal rain and river icons.	Icon set for explanation screens limited to literal pictograms with a word label.	Davies et al. (1998), <i>Applied Ergonomics</i> . Safety pictograms were poorly understood, "particularly those which were abstract". abstract Boedeker et al. (2023): concrete symbols with text helped people with Alzheimer's. Wu et al. (2022), <i>IJERPH</i> : older adults with MCI did best with realistic icons plus text. abstract
G79	Co-design with people with dementia C · Both New, manual marking	Explanation profiles should later be co-designed with users and carers.	Not buildable in this paper (no user study). Fits the limitations and future-work section.	Tielman et al. (2024). People with cognitive disabilities "should be involved throughout the development process of XAI". full text Atf & Lewis (2025). Recommends user-centred XAI "through participatory design". abstract Strong evidence, but it is a method, not a UI rule.
G80	Caregiver companion view C · Both Not marked in Excel Research Approach v2	Carer may see more detail; same prediction and invariants	The carer's view receives the same forecast with the exact figure and reasons, and shows which profile the user has	Mathur et al. (2026), <i>CHI</i> . AI explanations gave family members insight into older adults' daily lives; older adults imagined explanations shared with family. full text Miller, Whitlatch & Lyons (2016), <i>Dementia</i> . Decisions are commonly shared between people with dementia and family carers (36 studies). abstract
G81	Use familiar, real-world metaphors only C · Both New, manual marking	Likelihood as houses (icon array), not fan charts, error bars or probability curves.	Only icon arrays and the landmark picture are used to show uncertainty and level.	Galesic et al. (2009), <i>Health Psychology</i> . Icon arrays "increased accuracy of both low- and high-numeracy people", incl. adults aged 62–77. abstract Spiegelhalter et al. (2011), <i>Science</i> . Probabilities are "notoriously difficult to communicate" to lay audiences; graphic effectiveness depends on numeracy. abstract Ruginski et al. (2015), <i>CogSci</i> . Non-experts misread hurricane-cone uncertainty displays. abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G88	Literal language with visual support C · Both Not marked in Excel Research Approach v2	Pair each explanation element with a pictogram or photo	Every explanation line has a pictogram from one fixed set (rain, river rising, house, go-bag)	mixed evidence Fischer et al. (2023), <i>Weather, Climate & Society</i> . Warning messages with protective-action guidance in both text and graphic increased understanding, belief and deciding. abstract Houts et al. (2006), <i>Patient Educ Couns</i> . Pictures linked to text "markedly increase attention to and recall" of health information; low-literacy readers benefit most. abstract Counter: Hurtado et al. (2014), <i>J Intellect Disabil Res</i> . Text + pictures did no better than pictures only in an Easy Read leaflet. abstract
G107	Caregiver-mediated configuration C · Both Research Approach v2	Carer can set the profile with the user's consent	A carer-assisted setup path in onboarding, with a consent screen before the carer changes the profile; consent is stored	Mathur et al. (2026), <i>CHI</i> . Older adults saw AI explanations as shared with family and wanted to configure which explanations are shareable and at what detail. full text Largent et al. (2023), <i>J Am Geriatr Soc</i> : supported decision-making through a trusted supporter for people with MCI or mild dementia. Sinclair et al. (2019): families endorse it as reliance on others grows. abstract
G111	Calm colours for non-urgent states C · Both Not marked in Excel Research Approach v2	Colour follows risk category; a "possible" flood in 2 days is not red	The Forecast state uses a calm v2 colour token; red is kept for Danger and live alerts	mixed evidence Chapanis (1994), <i>Ergonomics</i> . White, yellow, orange and red were read as successively greater hazard. abstract Neussner (2021), <i>IJDRR</i> . Inconsistent colour codes in hazard warnings compromise understanding; recommends consistent yellow/orange/red levels. abstract Counter: Leonard (1999) and Mayhorn et al. (2017). Colour-hazard pairings are not always recognised; orange and yellow can read as dangerous as red. abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G114	Time-boxed tasks with countdowns C · Both Not marked in Excel Research Approach v2	Time-to-event as a concrete day or countdown with its window	A time chip reads "Thursday morning (about 2 days)"; preparation steps carry time estimates	Dallo et al. (2022), IJDRR. Hazard overviews with time and action indications significantly improved understanding of whether to act now (N=601). abstract Marx et al. (2022), JAACAP: meta-analysis of broad timing deficits in ADHD. Zheng et al. (2022): less accurate time perception in ADHD. abstract
G117	Definition tooltips C · Both Research Approach v2	Tap-to-define any technical term left in the explanation	Terms such as "danger level" or "upstream" are tappable and show a one-line definition	mixed evidence Bullock et al. (2019), Public Underst. Sci.. Jargon "impairs people's ability to process scientific information" (N=650). abstract Counter: Shulman et al. (2020), J Lang Soc Psychol. Jargon disrupted processing "even when definitions for the jargon terms are provided"; removing jargon beats defining it. abstract Tap-to-define tooltips themselves are untested.
G121	Local-language Easy Read track C · Both Research Approach v2	Easy Read explanation in Sinhala, Tamil, English	Every explanation template exists in Sinhala, Tamil and English Easy Read; the language comes from the profile	Trujillo-Falcón et al. (2022), BAMS. Significant language inequities in understanding translated watch/warning terms; translate meaning, not words. abstract Krocak et al. (2025), BAMS. Spanish speakers were less likely to receive, understand and respond to tornado warnings (N=14,341). abstract Sri Lanka context: Chandrasekara (2026) reports language gaps in early warning during Cyclone Ditwah (observational).

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G124	One decision at a time under stress C · Both Research Approach v2	Action-only while an alert is live; the "why" is deferred	When the forecast becomes a live alert (Layer 3), every profile shows hazard, action and a call button only; "Why?" returns when the alert ends	mixed evidence Mathur et al. (2026), CHI. For emergency alerts, older adults wanted "direct, task-focused and concise explanations"; elaboration added cognitive load under imagined stress. full text Starcke & Brand (2012): stress affects decision making. abstract Counter: Joslyn & LeClerc (2012), J Exp Psychol Appl: advice alone did not improve decisions, advice + uncertainty did. Wood et al. (2018): longer warnings cut information-seeking delay. Keep "Why?" one tap away. abstract
G125	Pre-rehearsed action plan C · Both Not marked in Excel Research Approach v2	Teach what "Possible" and "Likely" mean in calm time	A practice mode in calm time shows a sample "Possible" and "Likely" forecast and what to do for each	Budescu et al. (2009), Psychological Science. Readings of probability words deviated from the official meanings "even when the respondents had access to these guidelines"; pair words with numbers. abstract Wakui et al. (2017): caregivers of people with dementia were 36% less likely to have an evacuation plan. abstract No study shows that rehearsing warning terms in calm time improves later understanding.
G128	Multiple means of representation C · Both Not marked in Excel Research Approach v2	Same explanation as text, audio, symbols; modality is a profile attribute	Each explanation is available as text, read-aloud audio and pictograms; the profile picks the default modality	Szymanski et al. (2021), IUI. Lay users preferred visual explanations but did worse with them; hybrid visual + text explanations improved understanding. abstract Tait et al. (2012), Am J Med. Understanding was greater when the risk format matched the person's preferred format. abstract Nuance: Delaunay et al. (2025), PACM HCI. Graphical vs text mainly changed confidence, not comprehension. abstract

Indirect 9 guidelines

A source shows this property affects comprehension in general text or interfaces, or in people with dementia, MCI or autism. Applying it to explanations is our inference.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G15	Use simple tense and voice B · XAI New, manual marking	"Flooding is possible here", not "A flood event has been forecast with moderate certainty".	All explanation templates use active voice and present or simple future tense; checked in the screen spec.	mixed evidence Ferreira (2003), <i>Cognitive Psychology</i> . Passives "are frequently and systematically misinterpreted". abstract Sung et al. (2020), <i>JSLHR</i> . People with MCI had more difficulty with passive than active sentences. abstract Counter: Grossman & White-Devine (1998). Voice did not affect comprehension in Alzheimer's. No source found for tense. abstract
G17	Use literal language B · XAI New, manual marking	No metaphors for risk or confidence ("keep an eye on the river", "a perfect storm").	Fixed literal templates in Layer 2; the invariants check also rejects AI rewording with idioms.	Kalandadze et al. (2018), <i>Autism</i> . Meta-analysis: poorer figurative-language comprehension in autism ($g = -0.57$); not significant when matched on language ability. abstract Morsanyi et al. (2021). Medium accuracy disadvantage for idioms in autism. abstract No source on metaphor in AI explanations or warnings.
G38	Familiar metrics and units Need: Relate quantities to known references B · XAI Research Approach v2	River level as "knee-deep" or against a local landmark; local weekday and time	River level drawn against the gauge's local landmark (bridge mark) from stored reference levels; times in Sri Lanka time with the weekday in Sinhala, Tamil or English	Giambastiani et al. (2024), <i>preprint</i> . Familiar-object scale pictures gave significantly better comprehension than text analogies alone ($n=26$). abstract Chen et al. (2024), <i>CHI</i> . Analogies to familiar objects were rated as making data scale easier to grasp (self-rated). full text No flood-depth or landmark study found. Words alone ("knee-deep") may be weaker than a picture.
G23	Use white spacing C · Both New, manual marking	Space between explanation steps so each reads as its own unit.	1.5 line height and fixed gaps between steps on explanation screens.	Rello et al. (2015), <i>UAIS</i> . Larger text and character spacing sped reading for people with and without dyslexia. Reading speed, not comprehension. abstract

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G24	Foreground not obscured by background C · Both New, manual marking	No explanation text over the river photo or the icon array.	Text sits beside or under images, never on top.	Mitzner et al. (2006), <i>Human Factors</i> . Low contrast slowed older readers, but comprehension was not significantly affected. abstract Labarta et al. (2025), preprint. Legibility of XAI heatmaps changed how understandable they were (N=455). abstract
G27	Controls/content do not move unexpectedly C · Both New, manual marking	A revised forecast must not reflow the card under the user's finger; the change is announced first (G89).	Forecast card keeps fixed slots; new values arrive behind a "The forecast has changed" banner.	Bergmann et al. (2016). Older adults detected fewer changes, even in easy tasks. Veiel et al. (2006) : older adults slower to find changes. abstract Simons et al. (2000), <i>Perception</i> . Large gradual changes go unnoticed; supports announcing a revision. abstract
G49	Alternative content for complex info C · Both Research Approach v2	Easy Read version must carry the same facts (fidelity)	An Easy Read toggle swaps in the symbol-supported text built from the same forecast object; the invariants check runs on both versions	mixed evidence Hurtado et al. (2014), <i>J Intellect Disabil Res</i> . An Easy Read leaflet improved knowledge. abstract Counter: Wilson et al. (2026) : "empirical evidence of ERHI effectiveness was limited". Buell et al. (2019) : simplification alone gave no significant gain. abstract No source tests whether simplified versions keep the same facts (our fidelity point).
G84	Stigma-free language C · Both Research Approach v2	Name profiles by need, never by condition	Profile names ("Just tell me what to do" and others) appear in setup, settings and notifications; no condition names anywhere in the app	Kohl et al. (2025), <i>Gerontologist</i> . Stigma and fear lead people with dementia to conceal their diagnosis. abstract AboJabel et al. (2024), <i>BMC Med Ethics</i> : stigma is a barrier to assistive technology in dementia care. Kenny et al. (2016) , <i>Autism</i> : no single autism term is accepted by all. abstract No source on condition labels inside XAI tools.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G109	Repeat important information C · Both Not marked in Excel Research Approach v2	Periodic restatement reflects the latest prediction	While a Forecast or Warning is active, the home screen and repeat notifications read the latest forecast, never a cached one	mixed evidence Creighton et al. (2013) , <i>Int Psychogeriatr</i> . Review of 34 studies: spaced repetition can teach people with dementia new and known associations. abstract Han et al. (2017) , <i>Alzheimers Res Ther</i> . A tablet spaced-retrieval program modestly improved recall in MCI. abstract Counter: Ha et al. (2026) , <i>CHI</i> . Frequent repeated alerts led users to ignore or block them; restate on change, not on a timer. abstract

None found

3 guidelines

Marked as possibly relevant to XAI, but no source found links it to understanding an explanation, forecast or warning. Kept as our inference, to discuss.

ID	GUIDELINE	IMPLICATION FOR THE EXPLANATION	HOW TO ADAPT IT IN THE EWS	EVIDENCE
G20	Include symbols and letters necessary to decipher B · XAI New, manual marking	Sinhala and Tamil explanations render correctly, including numerals, % and dates. Supports G121.	Unicode and font test for every explanation string in all three languages.	Context only: Waidyanatha et al. (2007) . In Sri Lankan Sinhala/Tamil/English CAP alert trials, message integrity was compromised on small handsets. No comprehension measure. abstract
G42	Info to prepare for a task Need: Know what is coming and when B · XAI Not marked in Excel Research Approach v2	State expected timing and when the next forecast update arrives	Forecast screen shows "Next update: tonight at 6 pm" and what to prepare before the time window, read from forecast schedule metadata	No study on stating timing or the next forecast update was found.
G65	Focus Not Obscured (SC 2.4.11) B · XAI New, manual marking	"Why?" and "How sure?" are never hidden under a banner.	Layout test: explanation buttons stay visible when an alert banner appears.	No source links it to understanding. Suggest A.

Check before citing

- **NeuroAdaptX (Bunde, 2026)** is listed in the DESRIST 2026 proceedings, Part I (Springer, DOI 10.1007/978-3-032-28316-0_12), according to search results; the Springer and dblp pages themselves could not be opened from here. Its full text has not been read yet: check its claims (adaptive structure, modality and density; N=216 experiment) against the PDF before citing.
- Not opened: the OASIS CAP 1.2 specification itself (blocked here). Search results quoting its certainty values agree with the Research Approach: Likely = $p > \sim 50\%$, Possible = possible but not likely, $p \leq \sim 50\%$, Unlikely = $p \sim 0$. Read the spec once before citing. Also not opened: WCAG 1.3.1, 2.4.11 and 2.5.8 texts, Mileti & Sorensen (1990), Sutton & Kuligowski (2019), and the 2016 journal version of Ruginski et al.
- Sources linked to consensus.app were checked through that abstract record; their DOIs were not confirmed. Look up each DOI before adding it to the reference list.
- The catalogue cites Joddrell & Astell for G77 and G81, but their abstracts do not report an icon-recognition finding. Boedeker et al. (2023) and Wu et al. (2022) are the checked alternatives.
- Liao et al. (2020)'s third author is Sarah Miller, not Tim Miller.